FORMAL SMOOTHNESS TURNS THE DIFFERENTIALS TRIANGLE INTO A SHORT EXACT SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3417-3427 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3429-3440, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted A side only (spaces twin incomplete per ruling v3).

Editorial note. BORDERLINE (10th in the tally): the proof is the algebra restatement plus one exact citation — self-contained at the affine level; kept under the written-reduction reading, flagged for consistency with the pending BORDERLINE adjudication.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms, lemma triangle-differentials; Algebra, lemmas exact-sequence-differentials and ses-formally-smooth.

Source reference. Original lemma 00RS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3440–3440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 031K: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3440–3440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01UX: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 3440–3440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$, $g : Y \to S$ be morphisms of schemes. Assume f is formally smooth. Then*

$$0 \to f^*\Omega_{Y/S} \to \Omega_{X/S} \to \Omega_{X/Y} \to 0$$

(see Morphisms, Lemma 01UX) is short exact.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. The algebraic version of this lemma is the following: Given ring maps $A \to B \to C$ with $B \to C$ formally smooth, then the sequence

$$0 \to C \otimes_B \Omega_{B/A} \to \Omega_{C/A} \to \Omega_{C/B} \to 0$$

of Algebra, Lemma 00RS is exact. This is Algebra, Lemma 031K. □